

Galaxy Smart Troubleshooting Engine

Understand. Match. Validate. Resolve.

A hackathon-ready AI support experience for

Galaxy phone owners

Team: Byte4ce



THE PROBLEM

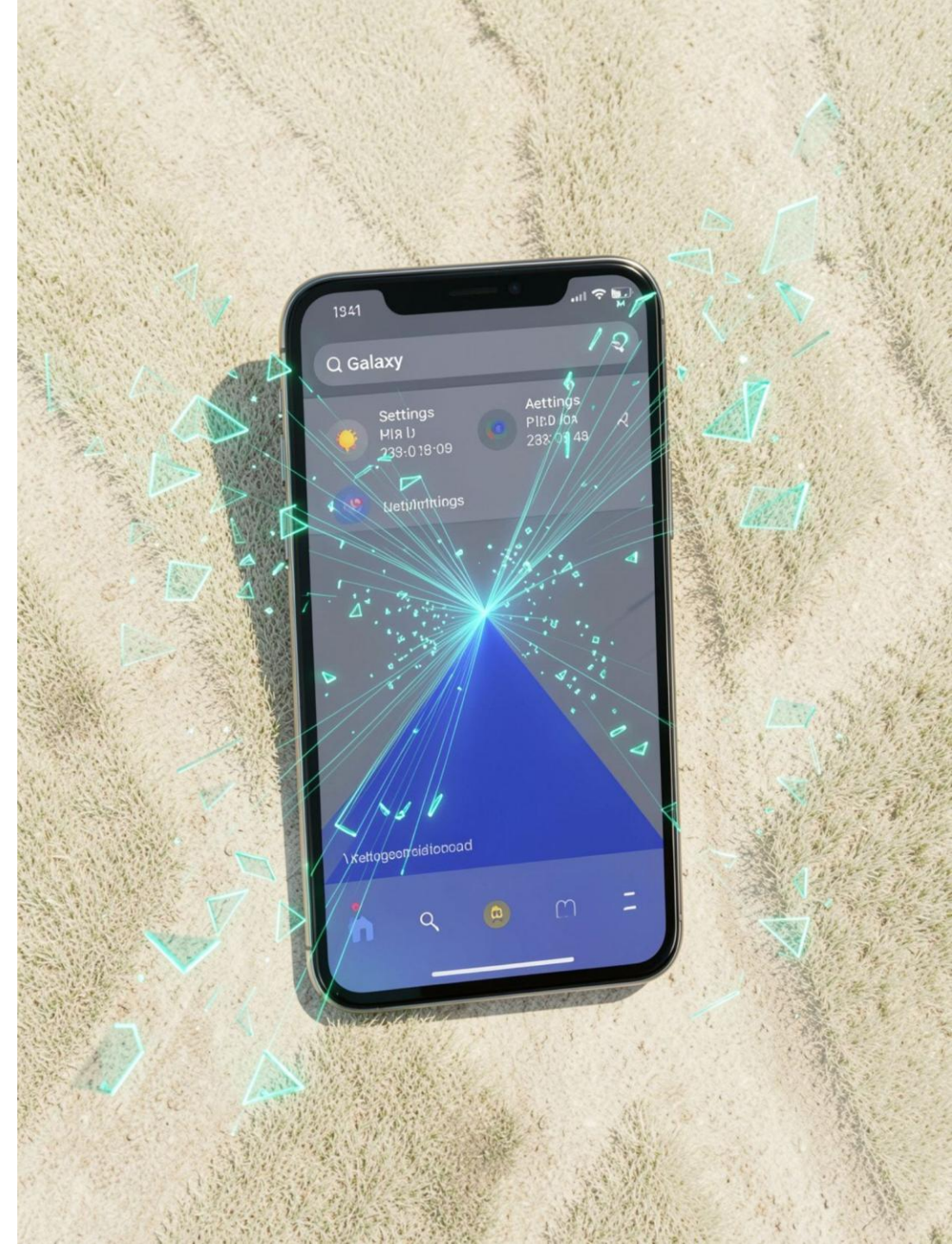
Phone support starts with messy human language

“My phone keeps doing that thing where it stops connecting after I leave home.”

Owners describe symptoms in their own words, language, and level of detail.

Traditional support: generic, link-heavy, and built around product menus.

Risk: unvalidated AI can invent settings paths, expose web links, or suggest risky steps too early.



THE SOLUTION

Our solution: a validated path from complaint to fix

Every answer moves through the same controlled journey.

01

Understand

Accept text, voice, screenshots, and multilingual complaints.

02

Match

Restate the issue, identify its domain, and ask when input is too vague.

03

Ground

Build a troubleshooting plan from One UI reference knowledge.

04

Validate

Map safe actions to approved settings screens.

05

Resolve

Deliver one-tap links—or escalate instead of guessing.

Architecture keeps the AI inside guardrails

Customer experience

React + TypeScript · text · voice · screenshot · language · device context

Express API

Understanding · cache · extraction · retrieval · validation · feedback loop

Server-side AI

Gemini grounded by One UI reference text

Control plane

Closed catalog of 36 entries · settings · cache · developer console

Validated response

Risk-labeled steps · approved deeplinks · confidence · escalation

Every request follows the same five-stage path

01

Input → Understand

Detect language, domain, intent, and injection attempts.

02

AI → Extract

Use grounded Gemini reasoning, or a rule-based fallback.

03

Cache

Reuse a validated answer for a semantically similar complaint.

04

Validation

Match actions with BM25 against approved catalog entries.

05

Final solution

Deliver risk-labeled steps, deeplinks, confidence, and escalation when needed.

Fast path · semantically similar complaint → validated cache answer

Cold path · new complaint → AI or rules → retrieval → validation

One validated output

Designed around how people actually ask for help



Multilingual

English, Hindi, Kannada, Tamil, and Hinglish understanding.



Voice input

Browser speech recognition for hands-free complaints.



Screenshot diagnosis

Show an error dialog or symptom instead of naming it.



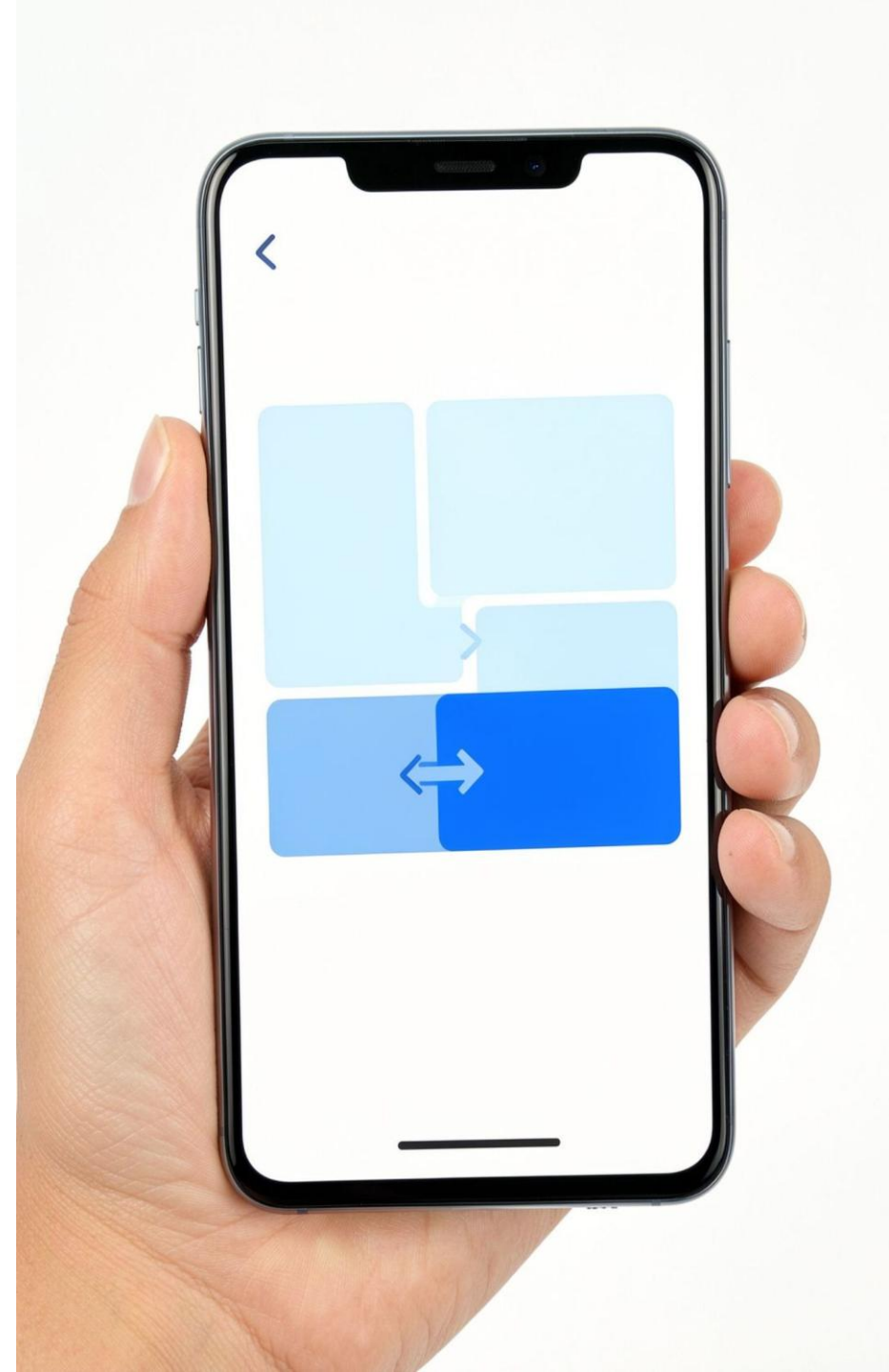
Semantic cache

Reworded complaints can return a fast-path answer.



One-tap settings + safety

Open the exact screen, order risk safely, and confirm critical actions.



THE STACK

The stack is practical, inspectable, and extensible

Frontend

React 19, TypeScript, Vite, Tailwind CSS.

Server

Express with a mobile-first customer view and developer console.

AI

Google Gemini API through the official SDK, configured server-side.

Retrieval

BM25 hybrid matching across a fixed settings catalog.

Operations

REST endpoints for troubleshooting, feedback, analytics, cache, catalog, and evaluation.

THE SAFETY MODEL

Safety and reliability are enforced in code

Guard the input. Ground the answer.
Validate the action. Escalate uncertainty.



Guard

Prompt injection: override attempts are refused before a model call.



Ground

Validation: URLs scrubbed; deeplinks must exist in catalog; titles, descriptions, and order repaired.



Validate

Offline fallback: rule-based understanding and extraction keep plans flowing when Gemini is unavailable.



Escalate

Confidence scoring: below 0.65 triggers a human escalation note; confident plans can enter the cache.

Risk controls

Physical hazards stop normal steps. Restart, safe mode, and reset stay last with confirmation.

From hackathon prototype to support platform

TODAY

Prototype

A clear customer experience plus an inspectable console for judges and engineers.

Closed-catalog grounding prevents invented settings paths and web redirects.

ADVANTAGE

Fast path

Semantic fast path improves repeat-query speed and reduces model dependency.

Validated answers can enter the cache.

NEXT

Platform

Replace lexical similarity with multilingual embeddings and persist cache and analytics.

Verify deeplinks on physical Galaxy devices, localize generated steps, and connect escalation to ticketing.

THE TAKEAWAY

Make support feel like a direct path to the right screen

Galaxy Smart Troubleshooting Engine turns vague complaints into grounded, validated, safe-to-follow actions.

- Understand the person, not just the product menu.
- Use AI where it helps, and rules where safety demands certainty.
- Give every fix a clear path, a confidence signal, and a human fallback.

Thank you — questions welcome.

